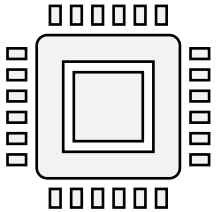
SAFESTEP NIGHT SYSTEM

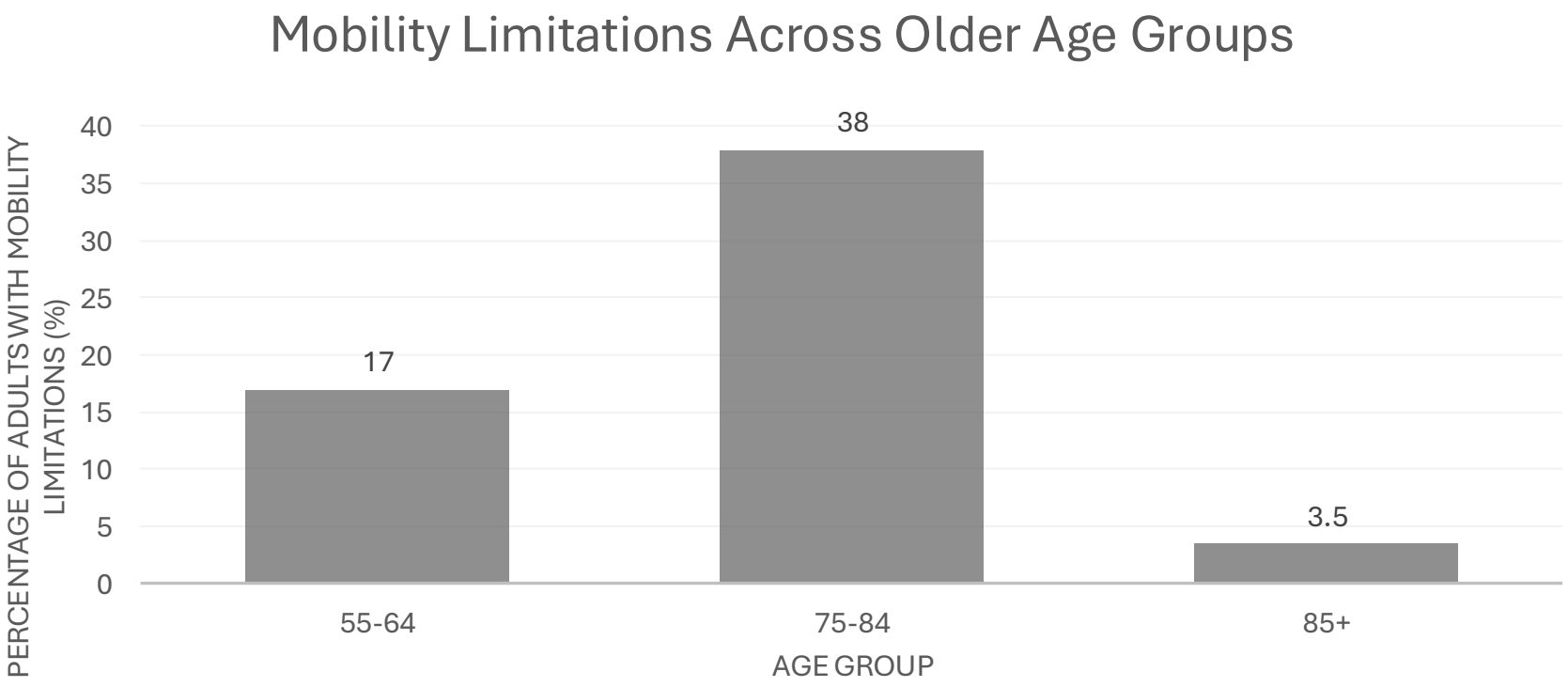
Remote-Assisted Night-Path Lighting for Seniors



Presented by Group 7

THE PROBLEM

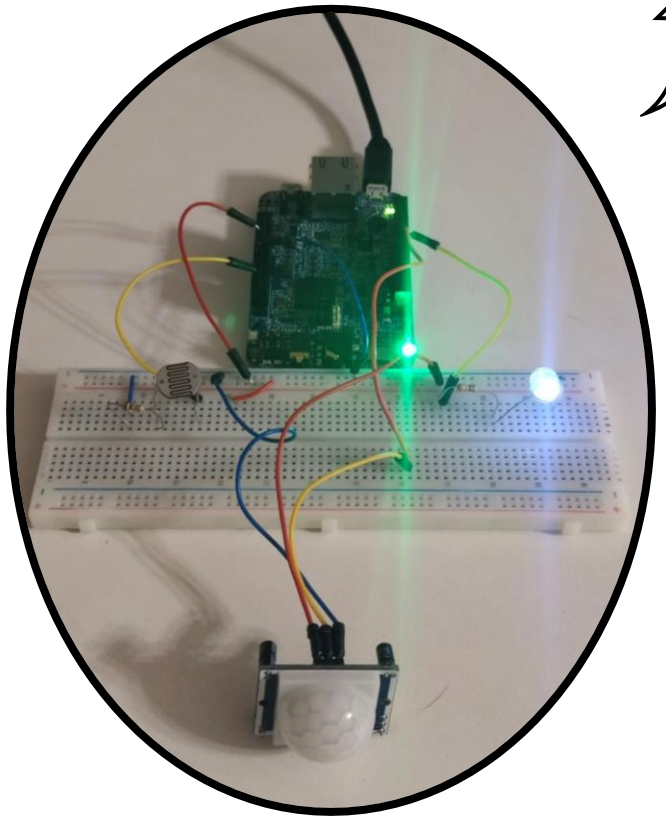
Night-time mobility is dangerous for seniors or people with vision/mobility impairments.
Key findings (2011 Federal Disability Report)



Up to 50% of falls occur at home during night-time movement, highlighting the increased risk during low-visibility conditions.

Proposed Solution

SafeStep Night System – Using FRDM-K66F



Detects darkness and motion
(LDR + PIR)

Caregiver lighting control
(brightness, timing, ON/OFF)

Adaptive LED Brightness
(PWM)

Logs night activity with timestamps
(RTC)

ML-based activity analysis
(Normal vs abnormal patterns)

Real-time system monitoring
(Connected dashboard)

EXISTING VS PROPOSED SOLUTION

Traditional Motion Light

- x Fixed timer
- x Fixed brightness
- x No monitoring

Safestep Night System

- ✓ Adaptive brightness & timing
- ✓ Remote caregiver control
- ✓ Live device state synchronization
- ✓ Timestamped activity logging
- ✓ ML-based safety analysis

SYSTEM HARDWARE

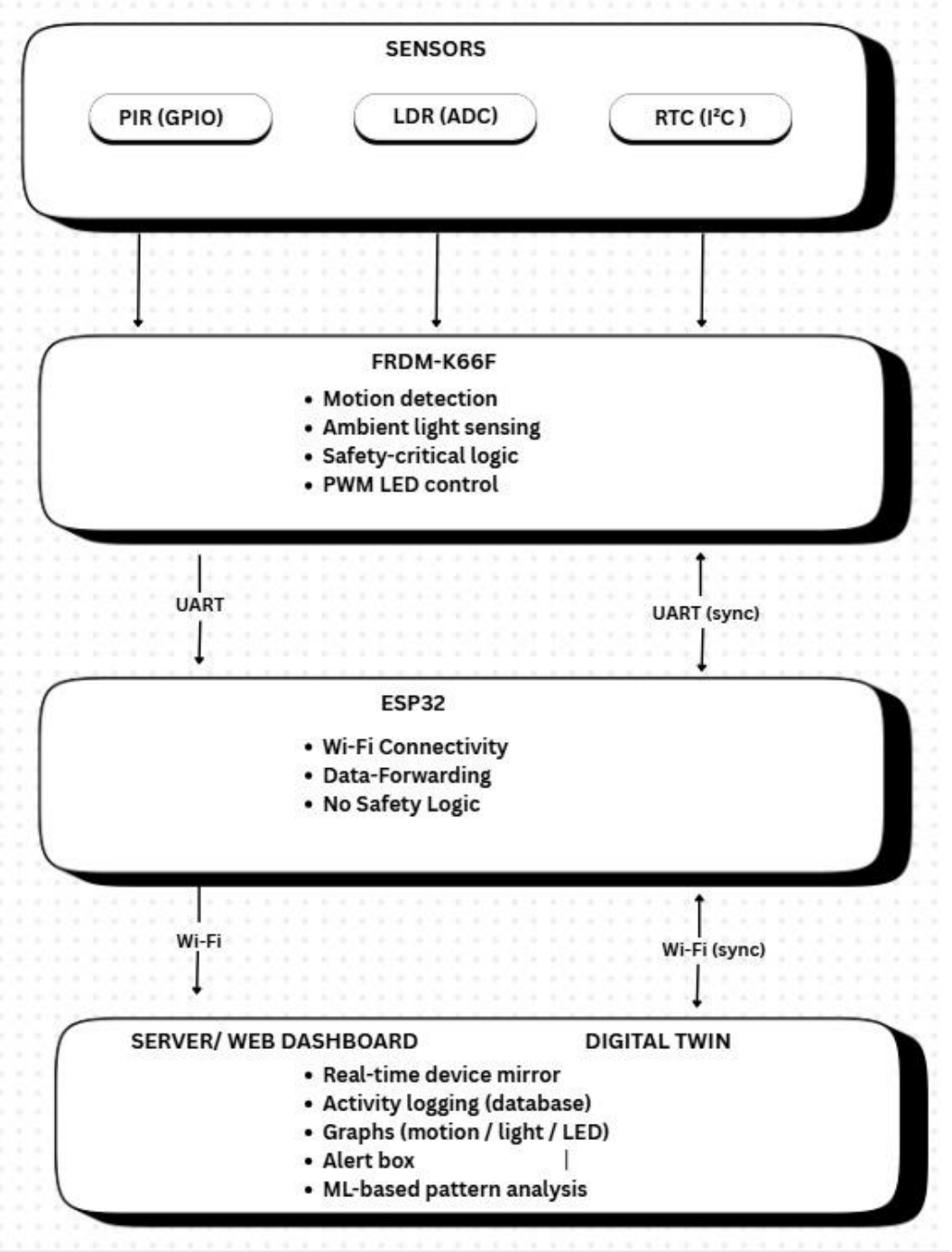
External Devices

LDR Light Sensor – senses ambient light level
PIR Motion Sensor – detects human movement
External LED – provides path illumination
RTC Module – provides real-time timestamps

MCU Peripherals & Communication Interfaces

ADC – reads analog light data
GPIO – reads inputs and drives outputs
PWM (FTM) – controls LED brightness
I^2C – communicates with the RTC
Ethernet – sends data to a server

System Architecture



PROJECT TIMELINE

Phase	Dates	Focus	Key Activities
Phase 1	Jan 24 – Feb 1	Proposal & Design	Problem definition, background research, system architecture, block diagram, UI wireframe
Phase 2	Feb 2 – Feb 16	Hardware & Embedded Logic	PIR, LDR, RTC integration; motion detection, PWM LED control
Phase 3	Feb 17 – Mar 2	Connectivity	UART communication with ESP32; Wi-Fi data transfer
Phase 4	Mar 3 – Mar 8	Milestone Prep	Dashboard prototype, data logging, milestone presentation + report
Phase 5	Mar 9 – Mar 22	ML & Digital Twin	Activity pattern analysis, alerts, two-way sync
Phase 6	Mar 23 – Mar 29	Demo & Final Report	Integration testing, live demo prep, GitHub submission, final report

⏱️ 40-Hour Work Scope

Activity	Hours
Background research & proposal prep	5
Architecture & block diagram	4
Sensor integration (PIR, LDR, RTC)	6
Embedded logic & PWM control	6
ESP32 UART + Wi-Fi communication	6
Dashboard & database logging	6
ML analysis & alert logic	4
Testing, demo & documentation	3
Total	**40 hours**

Thank you